National University of Colombia
Software Engineering II

Workshop 2

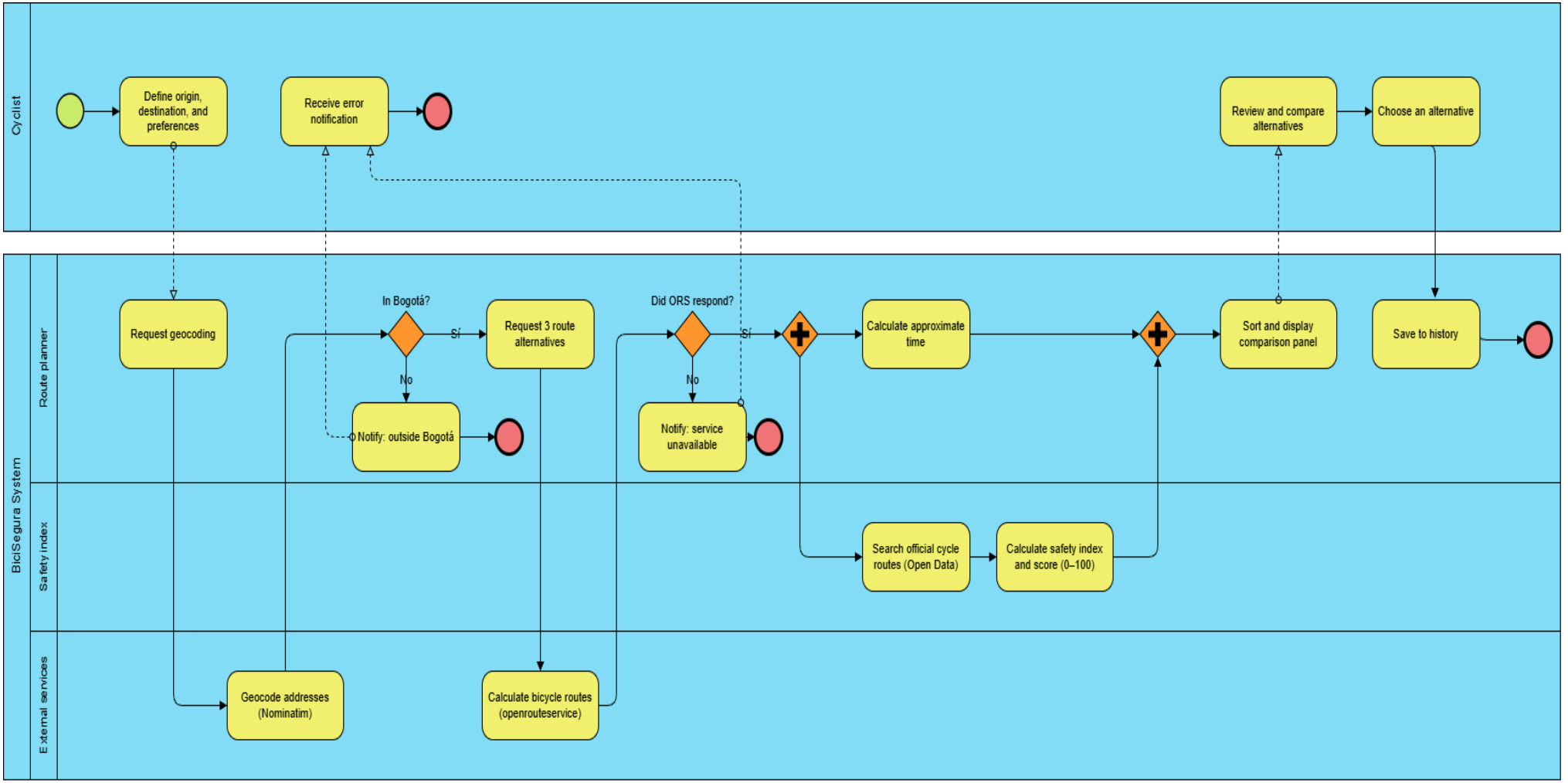
Design Artifacts and System Modeling

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6. Business Model Processes



Route request and comparison

Description

This process covers the moment a cyclist asks BiciSegura for a route. The cyclist defines an origin, a destination and preferences. The Route planner then asks the external geocoding service (Nominatim) to convert both addresses into coordinates. The system checks that the trip is inside Bogotá. If it is not, it notifies the cyclist and the process ends. Otherwise, the system requests three alternative bicycle routes from openrouteservice (ORS). If ORS does not respond, the cyclist is told the service is unavailable and the process ends.

When the routes arrive, two tasks run in parallel. The Route planner calculates the approximate travel time. At the same time, the Safety index lane calculates the percentage of each route that runs on official cycle routes (Open Data) and converts it into a safety score from 0 to 100. When both finish, the system sorts the alternatives and shows a comparison panel. The cyclist reviews the options and chooses one, and the system saves the choice in the cyclist's history.

Role in the application

- **Core use case:** it is the entry point to navigation. Everything that follows, such as following the route or recalculating it after a deviation, starts from the route the cyclist chooses here.
- **Main value of the product:** it shows safety next to distance and time, so the cyclist can pick a safer route and not only the fastest one.
- **Contained dependence on external services:** geocoding and routing come from third parties. The two error paths (outside Bogotá, service unavailable) make the system fail with a clear message instead of freezing.
- **Source of data for other features:** the saved history records which route type each user chooses, which feeds the administrator's statistics dashboard.

7. Architecture Diagram

BiciSegura follows a layered, client–server web architecture made up of three internal layers (presentation, application and data) that communicate with a set of external services. The diagram shows the components of each layer, the technologies used and the communication flows between them; the numbers in the figure correspond to the flow table below.

Presentation layer. The cyclist (from a smartphone browser) and the administrator (from a desktop browser) use a single-page application built with Vue.js. It contains the views of the system (login and registration, route planning and comparison, incident reporting, incidents and unsafe-zone map, active ride with alerts, and the administration panel), an HTTP client that consumes the REST API with the session token, and a Leaflet map that draws the base map, the incident and cycle-route layers, the heat map and the user's route and position. The interface is in Spanish and responsive from 360 px wide.

Application layer. The backend is a stateless REST API written in Python with Django. It checks the user's role on every endpoint and reads its business-rule parameters from a configuration file (SystemConfiguration), while API keys and credentials are kept only in environment variables. Its logic is organized in modules that correspond to the classes of the CRC cards.

Data layer. A PostgreSQL database with the PostGIS extension stores the 14 tables of the relational model (accounts, route searches and segments, community reports and moderation, navigation sessions, official cycle routes and the geocoding cache), including the geometry columns.

External services. The system uses OpenStreetMap for the base map, Nominatim for geocoding, openrouteservice for bicycle routes, the Datos Abiertos Bogotá cycle-route layer, the device GPS for the current location (with the user's permission) and an e-mail service for password-recovery links. If openrouteservice or Nominatim fail, the map and the incidents remain available and a clear message is shown.

Design decisions. The API is stateless so that it can be scaled and so that every request carries its own authentication token. The safety-index logic is separated from the routing provider so that openrouteservice could be replaced without changing the safety rules. Route alternatives are not persisted; only the route chosen by the user is saved. All external services are reached over HTTPS, and their usage limits (Nominatim, openrouteservice) and attribution requirements (OpenStreetMap, Secretaría Distrital de Movilidad) are respected.

BiciSegura — Software Architecture Diagram

Layers, components, technologies and communication flows (numbers refer to the flow table).

